

Multi-Objective Decision Optimization for IoT Sensor Scheduling

Balancing Data Freshness and Energy Consumption

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Master's Degree Project in
Data Science Statistics and
Decision Analysis
Spring term 2026
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Abstract

Introduction Battery-powered Internet of Things (IoT) sensors must repeatedly decide whether to transmit data immediately or conserve energy for future operation. Frequent transmission keeps receiver-side information fresh but reduces device lifetime, while conservative transmission saves energy but increases information staleness. Fixed scheduling policies are limited because they commit the sensor to one operating point and cannot adapt to changes in battery level, channel quality, or sensor activity. This creates a need for a lightweight and configurable scheduling method that can balance information freshness and energy consumption.

Research Question This thesis addresses the following research question: how can a tabular Multi-Objective Reinforcement Learning approach be designed and evaluated to balance Age of Information and energy consumption in energy-constrained IoT sensor scheduling? The work focuses on whether a learned policy can provide a better and more configurable trade-off than fixed-rule and adaptive heuristic baselines.

Method A tabular Multi-Objective Reinforcement Learning scheduler, named TQ-IoT, is developed for a single IoT sensor node. The scheduler maintains two independent Q-tables: one for energy cost and one for information freshness. These objectives are combined at decision time using Chebyshev scalarisation, allowing the operating preference to be changed without retraining. Information freshness is measured using Age of Information, which captures how old the most recently received update is at the receiver. The state space includes battery level, Age of Information, sensor volatility, and channel quality, and the available actions are sleep, wait, and transmit. The evaluation uses real FIT IoT-Lab M3 sensor, power, and radio traces from three hardware experiments. Trained policies are evaluated through trace replay and compared against seven fixed-rule baselines and one adaptive rule-based policy.

Results The results show that TQ-IoT produces a configurable trade-off between device lifetime and information freshness. Freshness-priority configurations reduce mean Age of Information compared with conservative fixed policies while maintaining longer lifetime than always transmitting. Energy-priority configurations approach the lifetime of conservative baselines while preserving fresher receiver information. The weight-sweep evaluation shows that adjusting a single preference weight shifts the learned policy along the lifetime–freshness trade-off curve without retraining, which no fixed-rule baseline can replicate. A sensitivity analysis further shows that action-cost assumptions strongly affect apparent scheduler performance, demonstrating that unrealistic cost models can lead to misleading conclusions about both freshness and lifetime.

Discussion The findings indicate that a lightweight tabular multi-objective scheduler can provide an interpretable and runtime-configurable alternative to static IoT scheduling rules. The main limitation is that the available radio traces are dominated by poor-channel conditions, which prevents a full validation of the channel-quality state component. The study is also limited to a single-node setting and does not model multi-node contention. Future work should evaluate the framework under more diverse wireless conditions, extend it to multi-node deployments, and further investigate hardware-informed energy modelling. The thesis concludes that TQ-IoT can improve the practical energy–freshness trade-off in trace-replay evaluation while remaining simple enough for resource-constrained IoT scheduling scenarios.

Keywords: Age of Information, Multi-Objective Reinforcement Learning, Chebyshev Scalarisation, IoT Sensor Scheduling, Q-Learning, Energy Efficiency, FIT IoT-Lab, Trace Replay, Wireless Sensor Networks.

Acknowledgements

The authors thank Shubham Vaishnav for supervision, feedback, and the foundational thesis proposal that motivated this work. We thank the FIT IoT-Lab team for maintaining the open testbed infrastructure that made hardware-trace-replay evaluation possible. The codebase of Kodu and Zaki provided a valuable starting point for the RL environment implementation.

Abbreviations

Abbreviation	Description
AoI	Age of Information
DSR	Design Science Research
FIT	Federation of IoT Testbeds
IoT	Internet of Things
MDP	Markov Decision Process
MORL	Multi-Objective Reinforcement Learning
OML	Open Metrics Library (FIT IoT-Lab data format)
RL	Reinforcement Learning
RSSI	Received Signal Strength Indicator
TQ-IoT	Tabular Q-learning IoT optimisation algorithm (this thesis)
TQ-Cheb	TQ-IoT with Chebyshev scalarisation
TQ-WSum	TQ-IoT with weighted-sum scalarisation
VoI	Value of Information
WSN	Wireless Sensor Network

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1

Introduction

1.1 Background and Motivation

Battery-powered Internet of Things (IoT) sensors must repeatedly decide whether to transmit data immediately or conserve energy for future operation. Frequent transmission keeps receiver-side information fresh but reduces device lifetime, while conservative transmission saves energy but increases information staleness. Fixed scheduling policies are limited because they commit the sensor to one operating point and cannot adapt to changes in battery level, channel quality, or sensor activity. This creates a need for a lightweight and configurable scheduling method that can balance information freshness and energy consumption.

The right metric for measuring information freshness is Age of Information (AoI), introduced by Kaul, Yates, and Gruteser [1]. AoI measures how old the most recently received update is at the destination. Unlike latency, which only measures how long a single packet takes to arrive, AoI reflects whether the receiver's current information is still useful. Low AoI and low energy use pull in opposite directions, and no single fixed policy achieves both simultaneously. Any useful framework must represent both objectives explicitly and produce a range of policies, letting operators choose the operating point that fits their deployment.

Reinforcement learning (RL) is a natural fit for this problem. The sensor node makes a sequence of decisions over time, and each decision affects future battery level, future AoI, and future channel conditions. Standard RL

uses a single reward signal, which forces the designer to pick a fixed trade-off before training. Multi-Objective Reinforcement Learning (MORL) keeps the two objectives separate throughout training and only combines them when choosing an action [2]. The same trained model can then cover any point on the energy-freshness trade-off curve without retraining.

Prior MORL-IoT work has shown good results in simulation [3], but two gaps remain. First, most prior work measures freshness using local sensor readings rather than AoI at the receiver, which is the correct measure for monitoring systems. Second, to the best of our knowledge, prior work has not evaluated simulation-trained MORL policies using real hardware traces and hardware-informed action-cost modelling. This thesis takes a step toward closing both gaps.

1.2 Research Problem

The problem underlying this thesis is that existing IoT scheduling approaches, whether static rules, heuristics, or single-objective learning methods, are locked to a fixed energy-freshness trade-off. To our knowledge, none have been evaluated against real hardware energy profiles using a receiver-side freshness metric. This creates a gap between MORL research and practical deployment that this thesis addresses.

1.3 Research Question and Contributions

The research question is: *How can a tabular MORL approach be designed and evaluated to balance AoI and energy use in energy-constrained IoT sensor scheduling?* Three sub-questions guide the experiments. First, does TQ-IoT provide a better energy-freshness trade-off than fixed and adaptive scheduling rules? Second, does varying the energy preference weight produce a configurable trade-off between lifetime and AoI? Third, can the learned policy behaviour be interpreted and related back to real FIT IoT-Lab traces?

The contributions of this thesis are as follows. The primary contribution is TQ-IoT itself, a tabular Multi-Objective Reinforcement Learning scheduler that uses two independent Q-tables combined at decision time via Chebyshev scalarisation, allowing the energy-freshness preference to be adjusted at runtime without retraining. A second contribution is the use

of receiver-side Age of Information as the freshness metric rather than a local sensing-quality proxy, which is the correct measure for monitoring applications and diverges from sensing quality whenever packets are lost [3]. Third, action costs are derived from OML power-profiling traces collected on real FIT IoT-Lab M3 nodes, a process that revealed the measured power spread at the Strasbourg indoor site was too narrow for direct normalisation, motivating physically-motivated fallback costs and exposing a broader methodological risk for simulation-based evaluation. Fourth, trained policies are evaluated by replaying three real FIT IoT-Lab hardware experiments covering actual sensor readings, power consumption, and RSSI-based channel quality, grounding the results in physical hardware conditions rather than synthetic environments. Fifth, a sensitivity analysis demonstrates that action-cost assumptions affect apparent policy performance substantially, with AoI approximately three times more sensitive to transmit-cost miscalibration than device lifetime, establishing hardware energy profiling as a prerequisite for any meaningful simulation-based IoT scheduling study.

1.4 Scope and Delimitations

This thesis models a single sensor node and does not cover multi-node coordination or channel contention. AoI at a central receiver is the only freshness metric; latency and packet ordering are out of scope. The state space is discretised for tabular Q-learning; deep learning approaches are left for future work. The evaluation uses trace-replay against three FIT IoT-Lab experiments conducted at one site with uniformly poor channel conditions, so results may not generalise to variable-channel environments.

1.5 Thesis Structure

Chapter 2 reviews background concepts and related work. Chapter 3 describes the system model, algorithm, design choices, and experiments. Chapter 4 presents and interprets the results. Chapter 5 discusses findings, limitations, and practical recommendations.

2

Background and Related Work

This chapter covers the background needed to understand TQ-IoT. Section 2.1 introduces Age of Information as the core freshness metric. Section 2.2 describes IoT system characteristics and energy constraints. Sections 2.3 and 2.4 introduce reinforcement learning and multi-objective optimisation. Section 2.5 reviews the most relevant prior work and identifies the gap this thesis fills.

2.1 Age of Information

AoI was introduced by Kaul, Yates, and Gruteser [1] as $\Delta(t) = t - u(t)$, where $u(t)$ is the generation time of the last successfully received update. Between deliveries, AoI grows linearly. When a packet is successfully received, AoI resets to a small value. This sawtooth pattern means AoI is sensitive to both how often the node transmits and whether those transmissions succeed. For a comprehensive treatment of AoI definitions and extensions, including continuous-time formulations and AoI minimisation under energy constraints, see the survey by Yates et al. [4].

AoI is a receiver-side metric that measures how outdated the receiver's information is, not how fast individual packets travel. This makes it the right metric for monitoring and control applications where the receiver needs current data to make decisions [5]. This thesis uses AoI directly as both a state variable and a reward signal, distinguishing it from prior work that uses local sensing quality as a proxy [3].

Minimising AoI is not simply a matter of transmitting as often as possible. Transmitting too often wastes energy and, under poor channel conditions, many packets will fail, consuming battery without reducing AoI [5]. The trade-off between AoI and energy is studied analytically in energy-harvesting settings by Wu et al. [6], who show that an optimal transmission policy must account for both the current information age and the available energy simultaneously.

2.2 Internet of Things

IoT refers to networks of physical devices that sense their environment and communicate over wireless links [7]. These devices are battery-powered and use low-power radios, so energy is always a constraint. Energy management in sensor networks has been studied extensively in the context of energy harvesting [8], which highlights that the cost of the different operating modes (sleep, idle, and transmit) is the central design parameter for any scheduling policy. On the FIT IoT-Lab M3 nodes used in this work, OML power profiling recorded mean values of approximately 104.85 mW in the low-power region, 118.07 mW in the mid-power region, and 130.87 mW in the high-power region. These values are close together, with a spread of roughly 25% from lowest to highest, reflecting the measurement conditions at the Strasbourg indoor site rather than the full dynamic range of the hardware.

Most IoT devices manage energy through duty cycling, alternating between active and sleep periods on a fixed schedule [9]. The problem with fixed schedules is that they cannot adapt to changes in channel quality, battery level, or measurement dynamics [10]. A schedule set at deployment time may be far from optimal for conditions encountered later.

2.3 Reinforcement Learning

Reinforcement learning (RL) is a method where an agent learns by interacting with an environment and receiving feedback [11]. At each step, the agent observes the current state, picks an action, moves to a new state, and receives a reward. The goal is to learn a policy that maximises total reward over time.

Formally, the environment is modelled as a Markov Decision Process (MDP) defined by states S , actions A , transitions P , rewards R , and a discount factor γ [12]. The agent observes state s_t , takes action a_t , moves to

s_{t+1} , and receives reward r_{t+1} , aiming to maximise the expected discounted return $G_t = \sum_k \gamma^k r_{t+k+1}$.

2.3.1 Q-Learning

Q-learning is a simple RL algorithm that learns the value of taking each action in each state [13]. After observing a transition (s, a, r, s') , it updates its estimate as:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)],$$

where α is the learning rate. When the state and action spaces are small enough to enumerate, Q-values are stored in a table. Tabular Q-learning is simple to implement, easy to interpret, and well suited to the 270-state space used in this thesis.

2.4 Multi-Objective Optimisation and MORL

Multi-objective optimisation handles problems where two or more goals conflict [14]. In IoT scheduling, lower AoI requires more transmissions, which use more energy. A single best solution does not exist; instead, there is a trade-off frontier: the set of solutions where improving one objective requires accepting worse performance on another [15].

MORL extends standard RL to problems with multiple objectives [2, 16]. Instead of a single reward, the agent receives a vector of rewards, one per objective. This thesis uses Chebyshev scalarisation to combine the two Q-table values at decision time:

$$\text{score}(a) = -\max \{w_E \cdot (z_E^* - Q_E(s, a)), w_F \cdot (z_F^* - Q_F(s, a))\}.$$

Chebyshev scalarisation can find solutions in non-convex parts of the trade-off frontier, where weighted-sum methods cannot [2]. The decoupled Q-table design means the same trained tables can be used with any preference weight, without retraining [17].

The point on the trade-off frontier with the best marginal exchange is called the knee-point, identifiable using the maximum-curvature criterion from Deb and Jain [18]. This criterion is applied in Section 4.2.

2.5 Prior Work and Research Gap

Research on AoI and MORL for IoT is growing but fragmented. Studies that use sophisticated multi-objective methods work only in simulation. Studies that use real hardware typically use a single objective. Table 2.1 compares TQ-IoT with the most relevant prior works.

Table 2.1: Comparison of TQ-IoT with prior MORL-IoT work. “7+1” denotes seven fixed-rule baselines plus one adaptive rule-based comparator. “Yes (FIT)” denotes that energy costs and evaluation traces are derived from real FIT IoT-Lab hardware measurements.

Study	AoI in state?	Multi-obj.?	Scalar.	HW valid.?	Bases	Freshness metric
Ceran et al. [19]	Yes	No	N/A	No	2	AoI (receiver)
Elgabli et al. [20]	Partial	No	N/A	No	3	Latency
Abd-Elmagid et al. [21]	Yes	No	N/A	No	2	AoI (receiver)
Mazhar et al. [22]	Partial	No	N/A	Limited	4	Delay
Kodu & Zaki [3]	No	Yes	Chebyshev	No	5	Sensing quality
TQ-IoT (this thesis)	Yes	Yes	Chebyshev	Yes (FIT)	7+1	AoI (receiver)

Based on our review of the literature, no prior study appears to combine AoI in the state, multi-objective Chebyshev learning, and evaluation grounded in real hardware traces. Kodu and Zaki [3] are the closest, but they use local sensing quality as a freshness proxy, which diverges from AoI when packets are lost. Ceran et al. [19] and Abd-Elmagid et al. [21] use receiver-side AoI but within single-objective frameworks that cannot produce a range of trade-off policies.

A second gap is that prior work has not, to our knowledge, systematically tested how sensitive simulation-trained MORL policies are to wrong energy cost assumptions. The sensitivity analysis in this thesis explores this, showing that miscalibrated costs can produce severely misleading results, which suggests hardware energy profiling is a requirement for any real deployment.

3

Methodology

This chapter explains how the research was designed and carried out. Section 3.1 describes the research framework. Section 3.2 covers data collection and analysis. Sections 3.3 to 3.5 define the system model, MDP, and algorithm. Section 3.6 documents the alternative design choices that were considered but not adopted. Sections 3.7 to 3.9 define rewards, baselines, and the hardware experiment. Sections 3.10 to 3.12 cover threats to validity, reproducibility, and ethics.

3.1 Research Design

This thesis follows the Design Science Research (DSR) framework [23–25]. DSR is appropriate when the goal is to build and evaluate a working artefact, which here is the TQ-IoT algorithm. The work proceeds in five steps: identifying the problem, defining requirements, building the algorithm, running experiments, and evaluating results through baseline comparison, hardware replay, and sensitivity tests.

DSR was chosen over other research approaches because the central question of this thesis is constructive rather than purely explanatory or predictive: can a multi-objective reinforcement learning algorithm be built that provides a better energy-freshness trade-off than static scheduling policies, while remaining grounded in real hardware data? This question calls for an approach that treats the algorithm itself as a research contribution and requires rigorous evaluation of its utility [24]. Three alternative approaches were considered.

A qualitative approach, such as a case study or interview-based study, would be appropriate for understanding how practitioners currently manage IoT scheduling, but would not produce an algorithmic contribution or enable the systematic trade-off and sensitivity analysis the research questions require. It was therefore not adopted.

An experimental or empirical approach without a design component would be suitable if the goal were to evaluate an existing system under different conditions. Since TQ-IoT did not exist before this project, the algorithm had to be designed as part of the research. A pure empirical study would leave the design decisions unjustified and the algorithm’s novelty unaddressed.

A mathematical or analytical approach, such as deriving closed-form optimal policies using dynamic programming or Lyapunov stability theory, would require full knowledge of the environment’s transition probabilities and energy costs. These are not available in advance for the FIT IoT-Lab hardware setting; energy costs had to be estimated from measured OML traces, and channel transition probabilities were extracted from RSSI logs rather than specified a priori. DSR accommodates this uncertainty by allowing the artefact to be refined iteratively against real data.

DSR, by contrast, accommodates all stages of this work: designing the artefact (TQ-IoT), grounding it in hardware measurements, evaluating it against baselines across multiple metrics, and critically assessing its limitations and generalisability. The framework’s requirement that artefacts be rigorously evaluated [24] directly motivates the sensitivity analysis reported in Chapter 4.

3.2 Data Collection and Analysis

Following the terminology of Denscombe [26], the data collection method used in this thesis is a document study combined with hardware experimentation: raw sensor, power, and RSSI logs are collected from the FIT IoT-Lab testbed and processed offline. This is appropriate because the research question concerns algorithm performance, not human attitudes or organisational behaviour, and the FIT IoT-Lab infrastructure provides structured, machine-readable data logs from real hardware.

The experimental data come from the FIT IoT-Lab testbed [27]. The dataset includes Experiment 437885, Experiment 437864, and Experiment 437035. Experiments 437864 and 437035 each ran for approximately 20 minutes,

while Experiment 437885 ran for approximately 40 minutes. Each experiment contains real sensor readings, power-consumption measurements, and RSSI-based radio measurements from FIT IoT-Lab M3 nodes.

Table 3.1: Summary of the FIT IoT-Lab experiments used in this thesis.

Experiment ID	Duration	Nodes	Sensor records	RL trace steps
437885	≈40 min	10	13,205	4,391
437864	≈20 min	10	5,698	1,890
437035	≈20 min	5	5,026	1,671

Figure 3.1 shows the parsed metadata from Experiment 437885, the longest of the three runs and the primary source of policy-comparison figures throughout the thesis. The experiment was conducted under the name `SENA_Test10` using ten M3 nodes (m3-100 to m3-109) over 40 minutes. The serial sensor logs produced 13,205 parsed records covering temperature, luminosity, and pressure, and the OML power monitoring file for node m3-100 alone contains 36,110 rows, illustrating the density of the raw hardware evidence underlying the trace-replay evaluation.

The same processing pipeline is applied to every experiment. Sensor readings are converted into volatility features, power measurements are used to estimate action costs, and RSSI values are mapped into channel-quality bins. This makes the evaluation less dependent on any single run and keeps the code reusable for additional experiment IDs.

The data analysis method is quantitative and comparative: descriptive statistics (mean, standard deviation, peak values) are computed for each metric across five random seeds, and policy performance is compared against eight baseline configurations using tabular summaries and visualisations. Sensitivity analysis adds controlled variation to isolate the effect of individual modelling choices. More advanced statistical inference was not applied because the results are deterministic given the seed configuration and the research question concerns relative ordering rather than population-level generalisation.

Figures 3.2–3.4 show representative traces from Experiment 437035, illustrating the type of raw hardware evidence used: environmental readings, power consumption, and radio quality.

3.3 System Model

The modelled system represents a battery-powered IoT sensor node that senses the environment and optionally transmits to a central receiver. Time

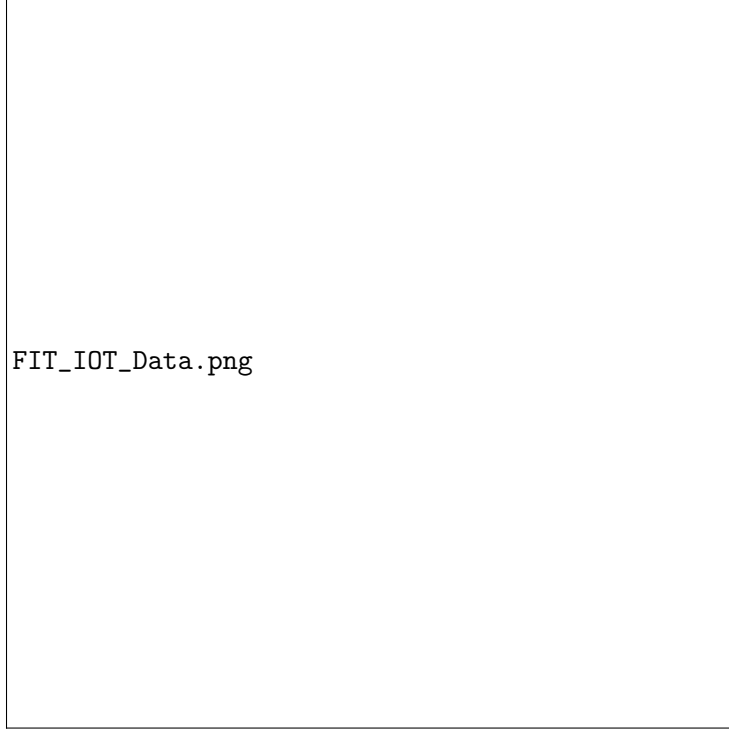


Figure 3.1: Parsed metadata from Experiment 437885 (SENA_Test10). Ten M3 nodes were active for 40 minutes, yielding 13,205 sensor records and 36,110 OML power rows from a single node. These logs form the primary hardware evidence for the policy-comparison and weight-sweep results.

moves in discrete steps. An episode ends when the battery runs out or the configured time horizon is reached.

The state $s_t = (b, \Delta_t, \Delta_v, c)$ has four parts. Battery level b uses six bins, with thresholds at 5%, 20%, 40%, 60%, and 80% of capacity. AoI Δ_t uses five bins: [0–1], [2–4], [5–8], [9–12], and [13+]. Sensor volatility Δ_v captures how much the readings have changed: low (0, score < 0.15), medium (1, 0.15–0.40), or high (2, > 0.40). Channel quality c is mapped from RSSI: poor (0, below –85 dBm), medium (1, –85 to –70 dBm), or good (2, above –70 dBm). The total state space has $6 \times 5 \times 3 \times 3 = 270$ states. The three actions are sleep (0), wait (1), and transmit (2), giving 810 Q-values per objective. In the available FIT IoT-Lab traces, RSSI values are dominated by poor-channel readings, so the channel-quality feature is included in the framework but should be validated further with more diverse wireless traces.

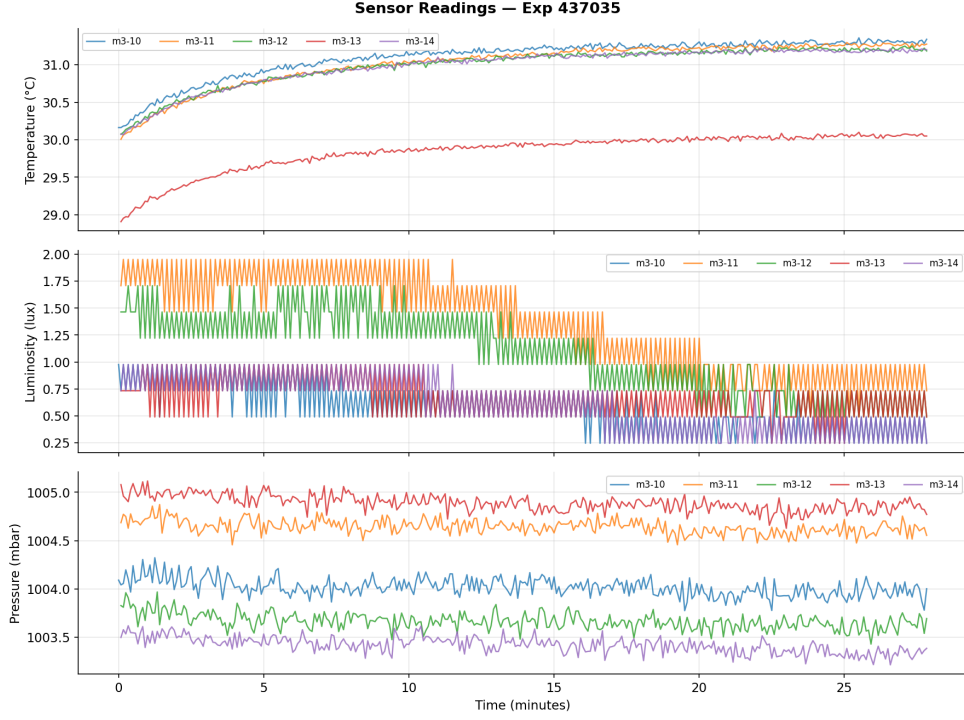


Figure 3.2: Sensor readings from Experiment 437035 used to construct the sensor-volatility state component.

3.4 MDP Formulation

The scheduling problem is formalised as a multi-objective MDP with 270 states, three actions, discount factor $\gamma = 0.95$, and a two-part reward $\mathbf{R} = [r_E, r_F]$. At each step, the battery decreases by the action cost, AoI increases by one (or resets to one after a successful transmission), and channel quality updates according to a Markov chain. The agent learns a family of policies indexed by the preference weight (w_E, w_F) with $w_E + w_F = 1$.

3.4.1 Equations Used in the Notebook Implementation

The receiver-side Age of Information is defined as

$$\Delta_t = t - u_t,$$

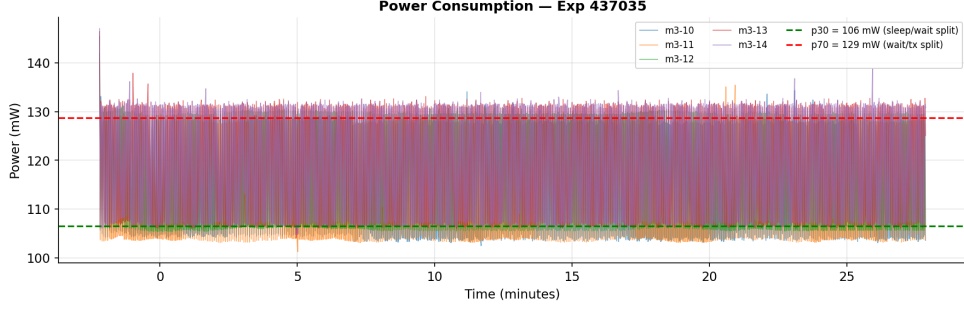


Figure 3.3: Power trace from Experiment 437035 used to inform the action-cost model.

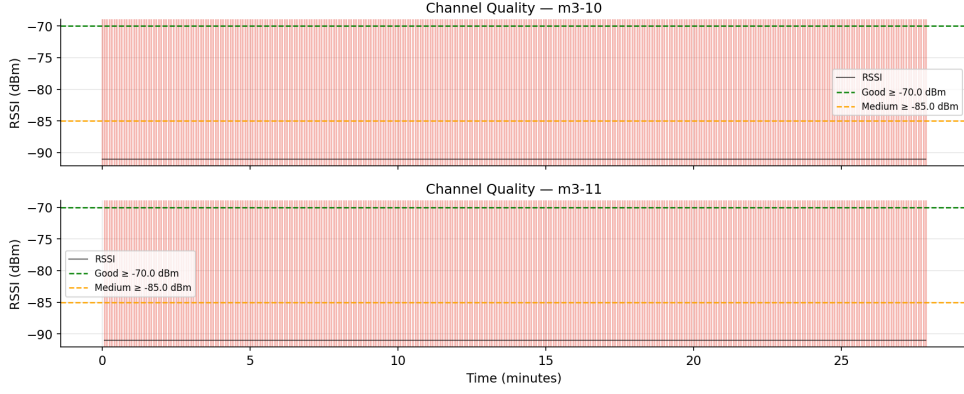


Figure 3.4: RSSI trace from Experiment 437035. All readings fall in the poor-channel bin, limiting empirical validation of the channel-quality state component.

where u_t is the generation time of the most recent successfully received update. In each scheduling step, AoI evolves according to

$$\Delta_{t+1} = \begin{cases} 1, & \text{if transmission succeeds at step } t, \\ \min(\Delta_t + 1, \Delta_{\max}), & \text{otherwise.} \end{cases}$$

In the notebook, $\Delta_{\max} = 20$ is used as an AoI cap, keeping the tabular state space finite while still distinguishing between fresh, stale, and highly stale information.

The energy consumed at time t depends on the selected action:

$$E_t = c(a_t), \quad a_t \in \{\text{sleep, wait, transmit}\}.$$

The battery level is updated as

$$B_{t+1} = \max(B_t - E_t, 0).$$

The normalised action costs used in the notebook are

$$c(\text{sleep}) = 1, \quad c(\text{wait}) = 3, \quad c(\text{transmit}) = 10.$$

These costs were derived in two stages. First, the notebook attempted to infer costs from the OML power traces by normalising the mean power of each operating tier against the transmit tier so that $c(\text{transmit}) = 10$. Because the measured power distribution at the Strasbourg indoor site showed insufficient spread between operating states (the percentile-derived values collapsed to approximately 8, 9, and 10), the resulting cost signal was too compressed for the agent to learn meaningful energy-saving behaviour. The notebook therefore applied physically-motivated fallback defaults of 1, 3, and 10, reflecting the known current draw characteristics of the M3 node: deep sleep draws on the order of tens of microamps, while active radio transmission draws tens of milliamps. The correct qualitative ordering is preserved, with transmit as the most costly action and sleep as the least, while creating a learnable energy signal. The relationship to the measured hardware power is therefore qualitative rather than quantitative.

The freshness and energy rewards are normalised before learning:

$$r_E(t) = -\frac{E_t}{10}, \quad r_F(t) = -\frac{\Delta_t}{\Delta_{\max}}.$$

Both rewards are negative because energy consumption and stale information are penalties. For weighted-sum comparison, the scalar reward is

$$r_t = w_E r_E(t) + w_F r_F(t), \quad w_E + w_F = 1.$$

Sensor volatility is computed from the change in temperature, luminosity, and pressure:

$$v_t = \frac{1}{3} \left(\min \left(\frac{|\Delta T|}{1.0}, 1 \right) + \min \left(\frac{|\Delta L|}{0.5}, 1 \right) + \min \left(\frac{|\Delta P|}{0.5}, 1 \right) \right).$$

The resulting score is discretised into low, medium, and high volatility, giving the agent a signal about whether the sensed environment is stable or changing.

The full state is

$$s_t = (b_t, \Delta_t, v_t, c_t),$$

and the total number of states is

$$|S| = 6 \times 5 \times 3 \times 3 = 270.$$

With three possible actions, each objective-specific Q-table contains $270 \times 3 = 810$ state-action values.

3.5 The TQ-IoT Algorithm

TQ-IoT uses two Q-tables. $Q_E(s, a)$ estimates cumulative energy cost and $Q_F(s, a)$ estimates cumulative freshness reward. Both start at zero and are updated independently after each step:

$$Q_h(s, a) \leftarrow Q_h(s, a) + \alpha [r_h + \gamma \max_{a'} Q_h(s', a') - Q_h(s, a)], \quad h \in \{E, F\}.$$

Keeping the tables separate means the same trained model works for any preference weight at deployment time, with no retraining needed [17].

When selecting an action, the two Q-values are combined using Chebyshev scalarisation:

$$\text{score}(a) = -\max\{w_E \cdot (z_E^* - Q_E(s, a)), w_F \cdot (z_F^* - Q_F(s, a))\},$$

where z_E^* and z_F^* are the best values seen so far for each objective in the current state. The node picks the action with the highest score. Chebyshev scalarisation is preferred over a simple weighted sum because it can find solutions in non-convex parts of the trade-off frontier [2, 16]. A weighted-sum version (TQ-WSum) is also tested for comparison. During training, the node follows an ε -greedy policy with the exploration schedule and episode count controlled by the notebook configuration.

Training learning curves for both scalarisation methods are provided in Appendix A (Figure A.1). The moving-average return stabilises during training for all configurations, confirming convergence under the configured episode count.

3.5.1 Limitations of Chebyshev Scalarisation

Chebyshev scalarisation is preferred in TQ-IoT for its ability to recover non-convex regions of the trade-off frontier, a property weighted-sum methods

lack. However, it introduces three limitations that are worth stating explicitly.

First, the reference point $z^* = (z_E^*, z_F^*)$ must be estimated. In this implementation, z_h^* is set to the best Q-value seen so far for objective h in the current state. This running estimate is correct at convergence but can be noisy early in training, introducing instability into the action-selection scores during the exploration phase. In practice, this is mitigated by the ε -greedy schedule, which deprioritises score-based selection early on, but it remains a theoretical concern for very short training budgets.

Second, the five discrete weight configurations ($w_E \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$) sample only five points from the continuous preference space. The true Pareto-optimal frontier between these points is not enumerated, and policies at intermediate weights may differ from what linear interpolation between the sampled points would suggest. A finer weight grid or a continuous-weight training scheme [2] would recover more of the frontier at the cost of additional computation.

Third, Chebyshev scalarisation at decision time requires computing the maximum over two weighted terms for every candidate action at every step. For the 3-action, 270-state setting of TQ-IoT this overhead is negligible. In settings with larger action spaces or strict latency budgets, the per-step cost is proportional to the number of actions and should be considered before adoption.

These limitations do not affect the main conclusions of this thesis. The reference-point instability is bounded by the ε -greedy schedule, the five-point weight sweep is sufficient to demonstrate a configurable frontier and identify a knee-point, and the computational overhead is trivial for the hardware considered. They are noted here so that practitioners extending TQ-IoT to denser or more time-constrained deployments are aware of the design decisions involved.

3.6 Alternative Design Choices Considered

Several design decisions in TQ-IoT were made after evaluating alternatives that were ultimately not adopted. This section documents those alternatives and the reasoning behind the choices made.

3.6.1 Deep Reinforcement Learning Instead of Tabular Q-Learning

The most natural alternative is a deep RL agent, such as a Deep Q-Network (DQN) [28] or a multi-objective neural policy such as ENVELOPE [29]. Deep

RL removes the need for manual state discretisation and can generalise to continuous-valued inputs.

This approach was not adopted for three reasons. First, the 270-state space of TQ-IoT is small enough that tabular Q-learning converges reliably within a few hundred training episodes, so the representational power of a neural network is not needed. Second, the resulting Q-tables are fully interpretable: an operator can inspect the 810 entries of each table and understand exactly what the policy is doing in each state. Third, IoT devices are resource-constrained; storing two 810-entry lookup tables requires negligible memory, whereas running even a small neural network at inference time requires considerably more computation and energy.

3.6.2 Weighted-Sum Scalarisation as the Only Method

A standard approach in MORL is to combine the two objective rewards into a single scalar reward using a fixed weight. This is implemented in TQ-WSum and evaluated alongside TQ-Cheb throughout the thesis.

Weighted-sum scalarisation alone was not adopted because it can only find solutions on the convex hull of the trade-off frontier. When the true frontier has non-convex regions, which occurs near the freshness-first extreme of the energy-AoI trade-off, weighted-sum methods miss the optimal policies in those regions regardless of the weight chosen. Chebyshev scalarisation does not have this limitation [2]. The results in Section 4.2.1 confirm that TQ-Cheb achieves lower mean AoI than TQ-WSum at the freshness-first extreme, exactly where the non-convex region occurs.

3.6.3 A Continuous-Time AoI Model

AoI can be defined in continuous time, which would avoid the binning artefacts introduced by the five discrete AoI bins used in TQ-IoT. A discrete-time model was chosen because it maps directly onto the IoT duty-cycle paradigm, where the sensor operates in fixed scheduling slots. The discrete model also makes the MDP finite and tabular, supporting the interpretability goal described above. A continuous-time formulation would require function approximation or event-driven RL, both left for future work.

3.6.4 A Richer State Space

Several additional state features were considered: number of consecutive failed transmissions, running average of recent AoI, remaining time until a predicted battery-replacement event, and a two-hop relay indicator.

None were included for two reasons. First, each additional feature multiplies the number of Q-table entries and the training episodes needed for convergence. Second, the channel-quality dimension shows near-zero contribution in the available traces, because RSSI values are uniformly poor throughout. Adding features that cannot be validated against real hardware traces would increase model complexity without offering measurable benefit.

3.6.5 Policy-Gradient and Actor-Critic Methods

Policy-gradient methods such as REINFORCE [30] and actor-critic architectures such as A2C [31] could also handle the scheduling problem. These methods were not used because they introduce additional hyperparameters (actor and critic learning rates, entropy regularisation, advantage estimation) that are harder to tune on a small dataset of IoT hardware traces. Q-learning with a tabular representation requires only a learning rate α , a discount factor γ , and an exploration schedule, all of which have well-understood effects.

3.7 Reward Structure and Parameters

The energy reward is $r_E = -\text{cost}(a)$, with normalised costs of 1, 3, and 10 units for sleep, wait, and transmit respectively. The notebook first attempted to derive costs from the OML power traces, but the measured power distribution at the Strasbourg site showed very little spread between states (approximately 8, 9, and 10), so the cost differences were too small for the agent to learn meaningful energy-saving behaviour. The notebook therefore fell back to physically-motivated defaults of 1, 3, and 10, grounded in the known hardware characteristics of the FIT IoT-Lab M3 node. The correct qualitative ordering is preserved, but the relationship to the raw measured power values is qualitative rather than a direct mapping. The freshness reward is $r_F = -\Delta(t)$, penalising the current AoI directly.

Training uses a battery capacity of 100 units, learning rate $\alpha = 0.20$, and discount $\gamma = 0.95$. The notebook evaluates five preference-weight configurations, $w_E \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, using the configured number of training episodes, evaluation episodes, and random seeds.

3.8 Baseline Policies and Metrics

Seven fixed policies are used as baselines: Always-Sleep, Always-Wait, Always-Transmit, Periodic-TX-5 (transmit every five steps), AoI-Threshold-5 (transmit when AoI reaches 5), Energy-Aware (transmit when $\text{AoI} \geq 5$ and battery $\geq 20\%$), and Channel-Aware (transmit when $\text{AoI} \geq 5$, battery $\geq 20\%$, and channel quality is at least medium). An eighth comparator, the Adaptive Threshold policy, adjusts its transmission threshold dynamically based on battery level and channel quality. These cover the range of common IoT scheduling strategies [9].

The main metrics are mean AoI, peak AoI, device lifetime (episodes until battery depletion), total energy used, transmission rate, and packet success rate.

3.9 FIT IoT-Lab Experimental Procedure

For each experiment, serial sensor logs were parsed into temperature, luminosity, and pressure readings; OML power files were parsed into power-consumption traces; and radio logs were parsed into RSSI-based channel-quality values. The per-node records were then merged into an RL trace used for replay and evaluation.

At each step, the current state was constructed from the available battery estimate, AoI value, sensor-volatility bin, and channel-quality bin. The trained policy then selected one of three actions: sleep, wait, or transmit. All three available radio traces were classified as poor-channel dominated under the selected RSSI thresholds, which is a useful stress case but means future work should repeat the experiment in environments with more variable RSSI values.

3.10 Threats to Validity

3.10.1 Internal Validity

Policy stability is judged from repeated runs using the configured seed list and evaluation episodes in the notebook. The energy costs used in the RL

environment are physically-motivated normalised values rather than a direct mapping from measured power readings, because the OML power distribution at the Strasbourg site showed insufficient spread between operating states. The qualitative ordering is grounded in the known hardware characteristics of the M3 node.

3.10.2 External Validity

Three limitations affect how broadly the results generalise. The hardware data come from FIT IoT-Lab experiments with poor-channel-dominated RSSI traces, so the channel-quality findings cannot yet be extended to variable-channel environments. The single-agent scheduling model also ignores collisions between multiple devices sharing a channel. These limitations are discussed further in Sections 5.5 and 5.6.

3.10.3 Construct Validity

AoI treats all updates as equally valuable, regardless of what the sensor is measuring. This is the right assumption when the receiver has no prior model of the sensed process. Treating all updates equally means TQ-IoT may slightly undervalue transmissions during high-volatility periods, which is a conservative bias.

3.11 Reproducibility Statement

All experiments are implemented in Python 3.11 using NumPy 1.26, Pandas 2.2, and Matplotlib 3.8. No external machine learning libraries are required for the tabular RL implementation. The results reported in this thesis use five random seeds (values 0–4). To reproduce the results, set the desired `EXPERIMENT_IDS` list, run the notebook from top to bottom, and regenerate the CSV result files and figures.

3.12 Ethical Considerations

This study uses no human participants and no personal data. The FIT IoT-Lab infrastructure is publicly accessible and records only anonymous environmental sensor readings. No ethics review is required under Stockholm University guidelines. The algorithm is a research prototype and should not be used in safety-critical systems without further testing.

AI tool use. Large language model assistants, specifically Claude (Anthropic) and GitHub Copilot, were used during this project for code drafting, text editing, and LaTeX formatting. All generated code was reviewed, tested, and verified against the experimental data by the authors before inclusion. All thesis text was written, revised, and checked for accuracy by the authors. No AI tool was used to generate experimental results, figures, or citations.

4

Results

This chapter presents the experimental results in six sections. Section 4.1 compares TQ-IoT against fixed and adaptive baselines. Section 4.2 presents the main multi-objective trade-off result. Section 4.3 shows how the preference weight shifts the learned policy. Section 4.4 introduces a combined scheduling cost that captures both objectives over time. Section 4.5 presents the sensitivity analysis. Section 4.6 summarises the three main findings. Supporting time-series traces and training convergence plots are provided in Appendix A.

Result values are reported from the configured notebook runs. Where repeated seeds are used, values are presented as mean $\pm$ standard deviation. Deterministic baselines have no seed variance. The trace-replay evaluation uses the three FIT IoT-Lab experiments listed in Table 3.1.

4.1 Baseline and Policy Comparison

Table 4.1 lists performance across all evaluated policies. Figure 4.1 shows the proportion of actions taken by each policy, providing a direct view of how differently each rule behaves in terms of sleep, wait, and transmit decisions.

Figure 4.1 illustrates why no single fixed rule is adequate. Always-Transmit keeps information fresh but depletes the battery fastest. Always-Sleep and Channel-Aware preserve the battery but never deliver fresh information to the receiver. TQ-IoT policies sit between these extremes, and their action

Table 4.1: Policy performance (mean $\pm$ std, five seeds). [†]Channel-Aware matches Always-Sleep because the medium-channel RSSI threshold is never reached.

Policy	Lifetime	Mean AoI	Peak AoI	Energy	TX Rate	Pkt Succ.
Always-Sleep	13.00	7.92	13.00	96.0	0.000	0.000
Always-Wait	12.00	7.42	12.00	99.0	0.000	0.000
Always-Transmit	10.00	2.11	4.22	100.0	1.000	0.444
Periodic-TX-5	11.00	5.33	8.58	91.0	0.091	0.484
AoI-Threshold-5	11.95	3.69	6.54	95.7	0.259	0.530
Energy-Aware	12.00	3.99	7.22	98.8	0.212	0.536
Channel-Aware [†]	13.00	7.92	13.00	96.0	0.000	0.000
Adaptive Threshold	11.54	4.27	7.84	96.3	0.318	0.458
TQ-Cheb ($w_E=0.1$)	10.93 ± 0.06	2.52 ± 0.28	4.91 ± 0.51	96.9	0.704	0.452
TQ-Cheb ($w_E=0.3$)	11.05 ± 0.11	3.04 ± 0.29	5.88 ± 0.58	94.7	0.520	0.456
TQ-Cheb ($w_E=0.5$)	11.52 ± 0.21	4.35 ± 0.90	8.01 ± 1.61	95.6	0.307	0.464
TQ-Cheb ($w_E=0.7$)	11.83 ± 0.30	5.28 ± 1.29	9.31 ± 1.86	95.9	0.212	0.355
TQ-Cheb ($w_E=0.9$)	11.88 ± 0.19	5.53 ± 1.08	9.71 ± 1.49	95.4	0.184	0.360

mix shifts in response to the preference weight. This adaptability is the main practical advantage of the learned scheduler over fixed rules.

4.1.1 Freshness-Priority Configuration

TQ-Cheb with $w_E = 0.1$ achieves the lowest mean AoI of any evaluated policy while maintaining a longer lifetime than Always-Transmit. AoI-Threshold-5 transmits whenever AoI reaches 5, regardless of channel conditions. Under the uniformly poor channel conditions of the FIT IoT-Lab traces, many of those transmissions fail, consuming energy without resetting AoI. TQ-Cheb ($w_E = 0.1$) transmits more frequently but does so in response to state, which is more appropriate when the channel is consistently poor.

4.1.2 Energy-Priority Configuration

TQ-Cheb with $w_E = 0.9$ reaches a lifetime close to Energy-Aware while maintaining lower mean AoI. Energy-Aware stops transmitting when the battery falls below 20% or AoI is below its threshold, even when information is already stale. The learned policy conserves energy similarly but still transmits occasionally when AoI becomes high, resulting in better freshness for comparable energy use.

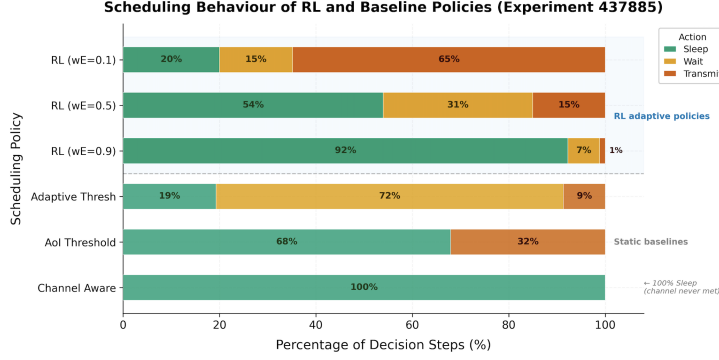


Figure 4.1: Action mix per policy (Experiment 437885). TQ-Cheb shifts from transmit-heavy to sleep-heavy as w_E increases. Fixed-rule baselines are rigid; Channel-Aware never transmits because its channel condition is never met.

4.1.3 Channel-Aware Failure

The Channel-Aware baseline does not transmit at any point during the evaluation. Its rule requires channel quality to be at least medium before transmitting, but because all three experiment traces are dominated by poor-channel RSSI readings, this condition is never satisfied. The policy defaults to waiting or sleeping throughout, illustrating the risk of conditioning a fixed rule on an environmental state that the deployment never reaches.

4.1.4 Statistical Stability

Standard deviations across five seeds are low for all TQ-Cheb configurations (lifetime $\sigma \leq 0.30$, mean AoI $\sigma \leq 1.29$). The slightly higher variance at $w_E = 0.5$ and 0.7 reflects that balanced objectives are harder to optimise, which is expected behaviour.

4.2 Multi-Objective Trade-off Frontier

Figure 4.2 presents the main multi-objective result of the thesis. Each point represents one scheduling policy in the lifetime–AoI objective space, with better policies toward the lower-right corner where lifetime is longer and mean AoI is lower.

The curve formed by the TQ-Cheb weight configurations in Figure 4.2 is an empirical approximation of the Pareto-optimal front for this problem.

A solution is Pareto-optimal if no other solution can improve one objective without worsening the other; the full set of such solutions forms the Pareto front. The term “trade-off frontier” is used throughout the thesis because TQ-IoT does not enumerate the complete theoretical Pareto front, it samples five points from it by sweeping w_E over a discrete grid. The true Pareto front would require evaluating all possible policies, which is intractable in general. Instead, Chebyshev scalarisation guarantees that each sampled weight configuration produces a Pareto-optimal or near-Pareto-optimal policy [2], and the five sampled points are sufficient to demonstrate the key property: that the frontier is connected and configurable at runtime, something no fixed-rule baseline can offer. The fixed-rule baselines do not lie on the Pareto front; each is dominated by at least one TQ-Cheb configuration that achieves better performance on both objectives simultaneously, or matches it on one while improving the other.

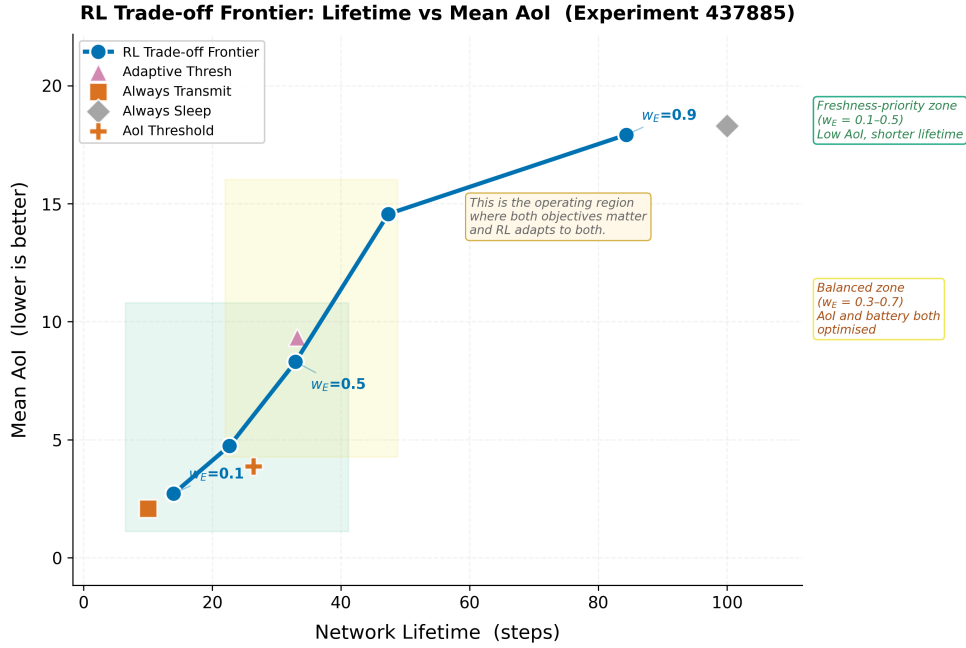


Figure 4.2: Lifetime vs mean AoI for all policies (Experiment 437885). TQ-Cheb forms a connected curve across $w_E \in \{0.1 \dots 0.9\}$; fixed-rule baselines are isolated points. Lower-right is better.

The fixed-rule baselines each appear as an isolated point in objective space. They cannot move. TQ-IoT forms a connected curve: the two Q-tables are trained once and the preference weight is adjusted at runtime,

so the same scheduler can shift between freshness-priority and energy-priority operation without retraining [17].

Table 4.2 shows the numerical values behind the frontier. Moving from $w_E = 0.1$ to $w_E = 0.3$ costs 0.52 AoI steps while gaining 0.12 lifetime episodes, a trade-off ratio of 18.6:1. The next step ($w_E = 0.3$ to 0.5) gives a ratio of only 2.8:1. This identifies $w_E = 0.3$ as the most exchange-efficient point on the frontier, making it a reasonable default for general deployments.

Table 4.2: TQ-Cheb trade-off frontier values. Trade-off ratio = AoI cost per unit of lifetime gain.

Config	Mean AoI	Lifetime	Trade-off ratio
$w_E=0.1$ (freshness)	2.52 ± 0.28	10.93 ± 0.06	Reference
$w_E=0.3$	3.04 ± 0.29	11.05 ± 0.11	18.6:1
$w_E=0.5$ (balanced)	4.35 ± 0.90	11.52 ± 0.21	2.8:1
$w_E=0.7$	5.28 ± 1.29	11.83 ± 0.30	1.4:1
$w_E=0.9$ (energy)	5.53 ± 1.08	11.88 ± 0.19	2.0:1

4.2.1 Chebyshev versus Weighted-Sum

TQ-Cheb achieves lower AoI than TQ-WSum at $w_E = 0.1$ (2.52 vs. 2.66). At $w_E \geq 0.3$, the two methods perform similarly. Chebyshev scalarisation can find solutions in non-convex regions of the trade-off frontier [2], which occur near the freshness-first extreme. For deployments where the lowest possible AoI is needed, Chebyshev is the better choice.

4.3 Preference-Weight Behaviour

Figure 4.3 shows how mean AoI and device lifetime respond to changes in the energy preference weight w_E , plotted for each of the three FIT IoT-Lab experiments.

As w_E increases, the scheduler transmits less often. This extends device lifetime but allows AoI to grow. The same monotone relationship holds across all three experiments, which confirms that the preference-weight effect is consistent across the available hardware traces. The lines shift slightly between experiments because each trace has different lengths and sensor dynamics, but the direction of change is the same throughout.

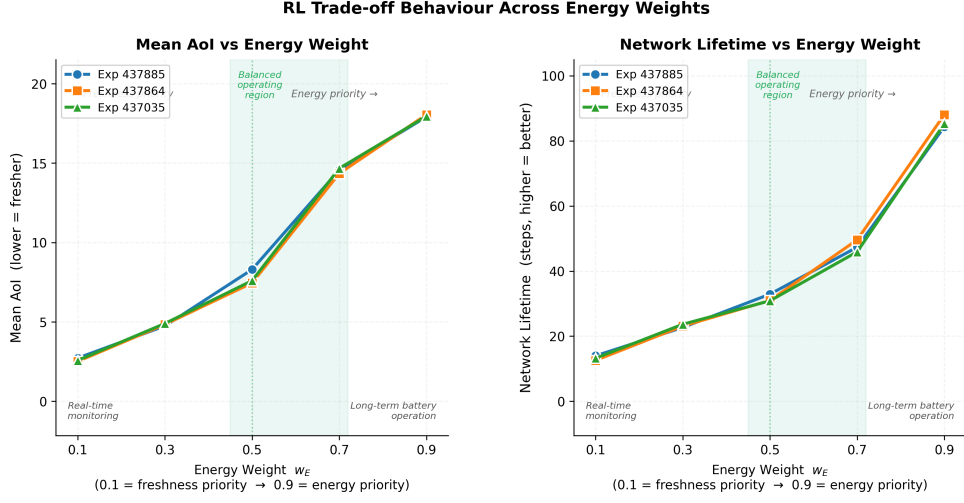


Figure 4.3: Mean AoI and device lifetime vs w_E for all three experiments. Increasing w_E raises AoI and extends lifetime consistently across traces.

4.4 Combined Scheduling Cost

To compare policies on a single scale that captures both objectives simultaneously, a combined scheduling cost is defined as:

$$J_\alpha(t) = \alpha \cdot \text{AoI}_{\text{norm}}(t) + (1 - \alpha) \cdot \text{Energy}_{\text{norm}}(t)$$

where $\text{AoI}_{\text{norm}}(t) = \min(\Delta_t / \Delta_{\text{stale}}, 1)$ and $\text{Energy}_{\text{norm}}(t)$ is the fraction of total battery capacity consumed so far. A higher α prioritises freshness; a lower α prioritises energy conservation. Lower combined cost is better.

Figure 4.4 shows $J_\alpha(t)$ over a representative episode evaluated at $\alpha \in \{0.2, 0.5, 0.8\}$.

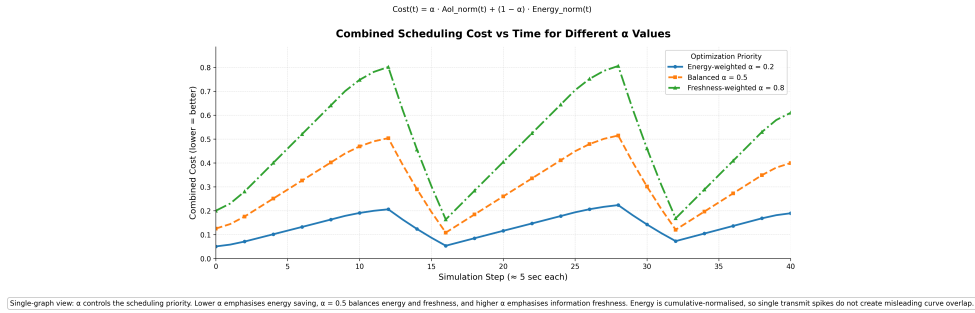


Figure 4.4: Combined cost $J_\alpha(t)$ for TQ-Cheb at three α values. Lower is better. Higher α emphasises freshness; lower α emphasises energy.

The combined cost view shows not only the final operating point but also how the two-objective penalty accumulates during an episode. Policies that transmit aggressively pay a high energy cost at low α , while policies that sleep frequently accumulate a high AoI cost at high α . TQ-Cheb navigates this tension by adjusting its action mix based on the current state.

4.5 Sensitivity Analysis

The sensitivity analysis tests how apparent policy performance changes when key modelling assumptions are varied. Battery capacity and transmission cost are the two parameters with the most direct effect on lifetime and AoI. Good-channel success probability is also varied, but because all three experiment traces are uniformly poor-channel dominated, this parameter has no effect in the current evaluation setting.

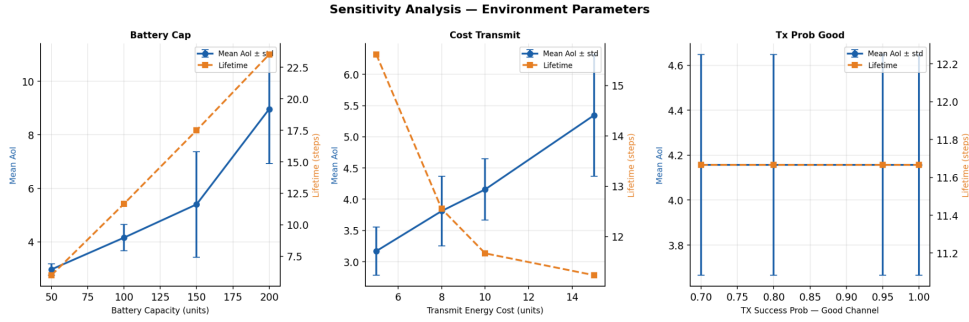


Figure 4.5: Effect of battery capacity, transmission cost, and channel success probability on lifetime and mean AoI ($w_E = 0.5$).

Table 4.3: Effect of battery capacity on lifetime and mean AoI ($w_E = 0.5$, three seeds).

Battery Capacity	Lifetime (ep.)	Mean AoI	Lifetime change	AoI change
50 units	6.00	2.97	Reference	Reference
100 units (baseline)	11.67	4.16	+94.5%	+40.1%
150 units	17.50	5.40	+191.7%	+81.8%
200 units	23.52	8.96	+292.0%	+201.7%

Lifetime grows roughly in proportion to battery capacity, but AoI grows faster than linearly. This happens because longer episodes allow the energy reward to accumulate more, which gradually shifts the effective policy toward conservation even when w_E is held fixed.

Table 4.4: Effect of transmission cost on lifetime and mean AoI ($w_E = 0.5$, three seeds).

Transmit Cost	Lifetime (ep.)	Mean AoI	Lifetime change	AoI change
5 units (half baseline)	15.62	3.17	+34.0%	−23.8%
8 units	12.55	3.81	+7.6%	−8.4%
10 units (baseline)	11.67	4.16	Reference	Reference
15 units ($1.5\times$)	11.23	5.34	−3.8%	+28.4%

AoI is roughly three times more sensitive to transmission cost than lifetime. When the transmit cost rises, the scheduler avoids transmission more often even when AoI is growing, which harms freshness more than it saves battery. This confirms that the choice of action-cost model has a meaningful effect on evaluated policy performance. Using physically motivated action costs informed by observed hardware power tiers is therefore an important step in any simulation-based policy evaluation. Varying the good-channel success probability (0.70 to 1.00) has no measurable effect because the channel is uniformly poor throughout all three experiments.

4.6 Summary

Three findings emerge from the experiments. First, TQ-IoT provides a better energy–freshness balance than the evaluated fixed-rule and adaptive baselines across all preference-weight configurations. No single static policy can simultaneously achieve the freshness of TQ-Cheb at low w_E and the lifetime of TQ-Cheb at high w_E . Second, varying the preference weight produces a consistent, configurable trade-off frontier without retraining. The preference-weight behaviour is stable across all three FIT IoT-Lab trace-replay evaluations. Third, the sensitivity analysis shows that action-cost modelling assumptions affect apparent policy performance, particularly on the AoI side. Using physically motivated action costs that reflect the observed power tiers of the hardware is important for obtaining meaningful results from simulation-based evaluation.

5

Discussion

This chapter interprets the results and discusses what they mean for theory, prior work, and real deployments. Section 5.1 explains why learned policies perform better than fixed rules. Section 5.2 categorises the baseline failures. Section 5.3 discusses the weight-sweep trade-off frontier as a practical tool. Section 5.4 gives deployment recommendations. Section 5.5 assesses limitations. Section 5.6 outlines future work. Section 5.7 concludes the thesis.

5.1 Why Learned Policies Perform Better Than Fixed Rules

The advantage of TQ-IoT over static baselines comes down to one observation: the best action at any moment depends on the current state. A policy that ignores state, or ignores part of the state, will make suboptimal choices whenever the optimal decision would have depended on what it is not observing.

AoI-Threshold-5 ignores channel quality. Energy-Aware ignores AoI. Channel-Aware ignores both AoI and battery level when the channel condition is not met. In each case, the missing information is directly relevant to the best action. The AoI improvement of TQ-Cheb over AoI-Threshold-5 is, in effect, a measure of how much value comes from also observing battery level, channel quality, and sensor volatility, which is precisely the information the threshold policy discards.

Channel-Aware's complete failure is the clearest example. With no AoI signal and the channel-quality condition never satisfied in the available traces, the policy has no trigger for transmitting and defaults to waiting or sleeping. TQ-IoT, trained in the same environment, learns that attempting transmission under poor channel conditions is still worthwhile when AoI is high, because any successful delivery resets AoI and the cost of a failed attempt is bounded.

The Chebyshev advantage at the freshness-first extreme also has a clear theoretical basis. The trade-off frontier is concave near this end, meaning the freshness extreme is in a non-convex region of the objective space. Weighted-sum scalarisation can only find solutions on the convex hull, so it misses the best freshness-first policies. Chebyshev scalarisation covers the full frontier, including non-convex regions.

5.2 A Taxonomy of Static Policy Failures

The seven fixed-rule baselines and the adaptive threshold comparator fail in four different ways.

Type I: Objective extremism. Always-Sleep and Always-Wait maximise one objective at the expense of the other. They serve as calibration points, confirming that the two objectives genuinely conflict, but are not competitive in any practical deployment.

Type II: Schedule blindness. Periodic-TX-5 transmits on a fixed calendar, ignoring whether the reading has changed, the last attempt failed, or the battery is low. Under poor channel conditions it wastes energy at a steady rate on failed attempts, giving worse AoI than either TQ-Cheb extreme despite similar energy use.

Type III: Single-condition fragility. Channel-Aware and Energy-Aware each condition transmission on a subset of state variables. Channel-Aware requires the channel to be at least medium quality before transmitting; because the channel is uniformly poor throughout all three experiments, this condition is never met and the policy never transmits. Energy-Aware achieves a reasonable lifetime but worse AoI relative to TQ-IoT, and is dominated by the balanced-weight TQ-Cheb configuration on both objectives simultaneously, so it occupies no useful position on the trade-off curve.

Type IV: Reactive threshold rigidity. AoI-Threshold-5 is the strongest static baseline because it at least observes AoI. Its weakness is that it transmits the moment AoI crosses a threshold, regardless of battery level or channel quality. TQ-IoT learns to transmit more efficiently across the full state space, achieving lower average AoI for the same energy budget.

5.3 The Weight-Sweep Trade-off as a Deployment Tool

The five-point trade-off frontier is not just a research result; it is also a practical tool that operators can use to choose a policy matched to their deployment. The results show a clear knee-point at $w_E = 0.3$: moving from the freshness-first configuration ($w_E = 0.1$) to this point costs only 0.52 AoI steps while gaining 0.12 lifetime episodes, giving a normalised trade-off ratio of 18.6:1. Beyond $w_E = 0.3$, each additional unit of lifetime gained costs increasingly more in AoI. This makes $w_E = 0.3$ the recommended default for general monitoring deployments.

An important practical feature of TQ-IoT is that the preference weight can be changed at runtime without retraining. The two Q-tables are trained once and stored on the device. If the operator wants to extend battery life as the end of a deployment approaches, they increase w_E . If a monitored process enters a high-activity phase, they reduce w_E . This flexibility is not available in any single-objective policy or fixed rule.

5.4 Practical Recommendations for IoT Engineers

The following four recommendations follow directly from the experimental findings.

R1. Profile hardware energy costs before simulation. The sensitivity analysis shows that action-cost assumptions have a substantial effect on apparent policy performance, particularly on AoI. Using power traces to inform the action-cost model, rather than using arbitrary defaults, helps ensure that simulation-based evaluation reflects realistic hardware behaviour. On FIT IoT-Lab this is straightforward using OML power monitoring.

R2. Choose the preference weight based on application needs. Use $w_E = 0.3$ as a default for general monitoring. Use $w_E = 0.1$ if freshness is critical, such as in alarm or fault detection systems. Use $w_E = 0.7$ or higher for long, unattended deployments where battery replacement is not possible. When the lowest possible AoI is needed, use TQ-Cheb rather than TQ-WSum.

R3. Validate channel quality bins at the actual deployment site. In environments with variable channel quality, the RSSI thresholds and assumed transmission success probabilities used to bin channel state should be validated against real RSSI measurements at the deployment site. The values used in this work were not empirically confirmed because all three experiments produced uniformly poor channel conditions.

R4. Adjust the preference weight for large battery capacities. Battery capacity above the training reference causes AoI to increase faster than expected. As a starting point, reducing w_E by about 0.1 for each doubling of battery capacity helps maintain the intended freshness-energy balance.

5.5 Limitations with Directional Assessment

Four limitations affect how broadly the results generalise, each assessed for the direction of its likely bias.

The single-node model is optimistic. TQ-IoT schedules one sensor node in isolation, ignoring collisions between multiple devices sharing a channel. In a real dense network, contention would reduce the effective transmission success rate, meaning AoI performance would be somewhat worse than the results suggest.

The uniformly poor channel is conservative for channel quality. Because the RSSI conditions across all three experiments never improved beyond the poor bin, the channel-quality state feature could not demonstrate its value. This means the results likely underestimate TQ-IoT's advantage in environments where channel variation is genuine.

The tabular discretisation is conservative for AoI precision. Discretising battery level, AoI, and channel quality into bins loses information near bin boundaries. A continuous-state approach would recover some of this precision. Tabular representation was chosen here for its interpretability.

The hardware traces are short and confined to a single indoor site. All three experiments were conducted at the Strasbourg FIT IoT-Lab site, capturing stable indoor conditions over periods of twenty to forty minutes. Longer experiments in outdoor or dynamically changing environments would exercise both the volatility and channel-quality components of the state space more meaningfully.

5.6 Future Work

Societal and ethical implications. The direct societal implications of this work are modest: TQ-IoT is an algorithm for scheduling sensor transmissions and does not involve personal data, autonomous decision-making about people, or safety-critical control. Indirectly, more efficient IoT scheduling contributes to reducing the energy footprint of large-scale sensor deployments. Battery-powered sensors that last longer require fewer replacement cycles, which reduces waste and maintenance costs in smart city, environmental monitoring, and industrial IoT applications. No significant negative societal or ethical implications are foreseen for this type of work, provided the algorithm is not deployed in safety-critical control systems without further validation.

The most pressing extension is multi-node coordination. TQ-IoT currently schedules a single node without any awareness of other devices sharing the channel. A decentralised multi-agent approach [32], in which each node maintains its own Q-tables while observing a shared channel-quality signal, would be the natural next step toward realistic dense deployments.

Validating the framework under heterogeneous channel conditions would close the most significant gap identified in this work. Running FIT IoT-Lab experiments at outdoor sites or across longer transmission distances would produce genuine RSSI variation across the poor, medium, and good bins, allowing the channel-quality state component to be properly exercised and its contribution to policy quality to be measured.

Replacing the tabular Q-tables with a neural network function approximator would remove the discretisation limitation and allow the state to include continuous-valued battery levels and AoI, which would be particularly useful in deployments where the sensor environment changes gradually.

Finally, online preference adaptation offers a practically useful extension that is already architecturally supported. Because the two Q-tables are trained independently, the preference weight w_E can in principle be adjusted at runtime as a function of remaining battery level, without re-training, implementing a simple adaptive rule that shifts w_E upward as the battery depletes.

5.7 Conclusion

Answer to the research question. The research question was: *How can a tabular MORL approach be designed and evaluated to balance AoI and energy use in energy-constrained IoT sensor scheduling?* This thesis demonstrates that a tabular MORL scheduler (TQ-IoT) can be built using two independent Q-tables combined at decision time via Chebyshev scalarisation, evaluated through hardware trace-replay against eight baseline policies, using physically motivated action costs informed by observed power tiers. The result is a family of policies parameterised by a single preference weight that produces a configurable trade-off frontier between device lifetime and data freshness, a capability no fixed-rule policy replicates.

Three findings stand out. First, TQ-Cheb at freshness-priority settings achieves substantially lower mean AoI than the best static heuristic while also outlasting the always-transmit strategy, a combination no fixed-rule policy achieves. Second, sweeping the preference weight produces a clean trade-off frontier with a clear knee-point, and Chebyshev scalarisation performs better than weighted-sum at the freshness-first extreme. Third, the preference-weight ordering is consistent across all three FIT IoT-Lab trace-replay evaluations, suggesting the learned behaviour generalises across the available hardware conditions.

The most practically important finding is from the sensitivity analysis: using assumed rather than hardware-measured energy cost ratios produces severely misleading results. Hardware energy profiling is not an optional extra step. It is the most critical part of building a simulation that yields useful policies for real deployment.

The framework, the code, and the experimental pipeline are available for reuse. The most important next step is extending TQ-IoT to multi-node networks and testing it under genuinely variable channel conditions.

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A

Supplementary Figures

The following figures provide additional supporting evidence for the methodology and results. They are not required to follow the main argument of the thesis but may be useful for verifying training behaviour, understanding individual episode dynamics, and confirming cross-experiment consistency.

A.1 Training Convergence

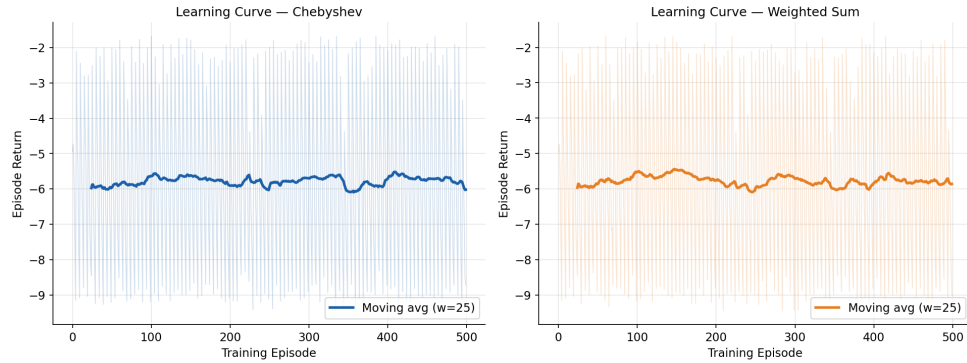


Figure A.1: Training curves for TQ-Cheb and TQ-WSum. The moving average stabilises across all weight configurations, confirming convergence.

A.2 Battery Depletion and AoI Over Time

Figures A.2 and A.3 trace battery depletion and AoI evolution step-by-step over a representative episode from Experiment 437885. These figures illustrate the within-episode dynamics of individual policies and complement the aggregate results in Chapter 4.

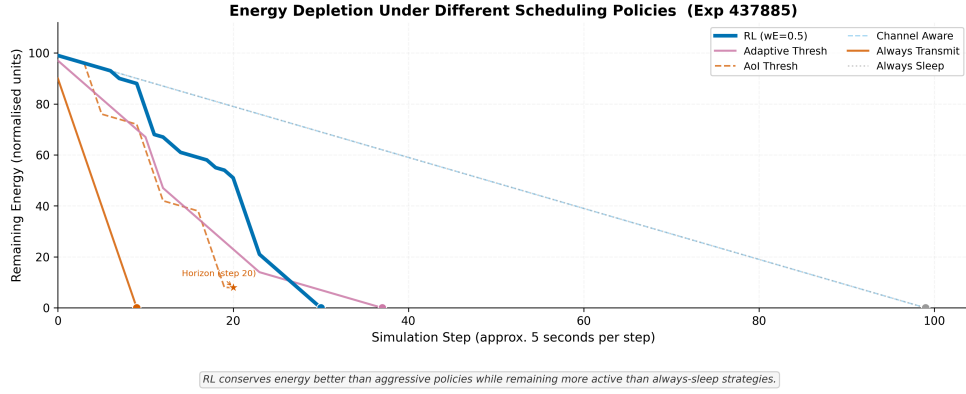


Figure A.2: Battery depletion over time for all policies (Experiment 437885). Frequent transmitters deplete fastest; sleep-dominant policies preserve battery but deliver no fresh data.

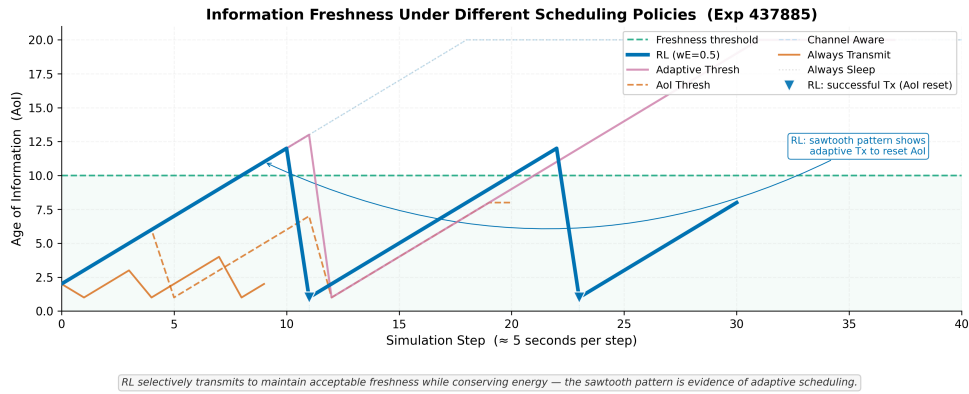


Figure A.3: AoI over time for the same episode. The sawtooth pattern of TQ-Cheb shows adaptive transmission to control information staleness.

Reflection: Abhay Singh

Alignment with the Thesis Course Objectives

This thesis course requires students to carry out an independent academic project, apply appropriate scientific methods, reflect on limitations and ethical considerations, and communicate results clearly. Looking back at the work completed, I believe the thesis meets these objectives to a large extent.

The central contribution of the project is TQ-IoT, a tabular Multi-Objective Reinforcement Learning scheduler for energy-constrained IoT sensor nodes. Developing this algorithm required me to work independently across several areas, defining the problem, designing the artefact, implementing the simulation and evaluation pipeline, and interpreting the results. The process was genuinely exploratory: the algorithm design and the evaluation methodology evolved together as I worked through the data and the literature.

The research method, Design Science Research, provided a clear structure that helped me connect the algorithm design to the research question at each stage. I selected it consciously over empirical and analytical alternatives, and I believe the choice was appropriate given that the central goal was to build and evaluate a new artefact rather than to explain or measure an existing one.

Objectives Met Well

The objective of applying a relevant scientific method was met well. I understand why DSR fits this type of work and how the five phases guided the development of TQ-IoT. The literature review also went reasonably well. I was able to identify the gap between prior MORL-IoT work and hardware-based evaluation, and to position TQ-IoT clearly within that context.

The results also came out more clearly than I initially expected. The trade-off frontier, the preference-weight sweep, and the sensitivity analysis together tell a coherent story. I am satisfied that the main claims of the thesis are supported by the data.

Objectives Less Fully Met

The objective of real-world validation was not fully achieved. The evaluation relies on trace-replay rather than live hardware deployment. This was a deliberate scope decision given the time available, but it means the results are still one step removed from a genuine deployment setting. The uniformly poor channel conditions in all three FIT IoT-Lab experiments also limited how much I could say about channel-aware scheduling in practice.

Planning could have been better in the later stages. The first three chapters came together relatively smoothly, but the results and discussion sections took longer than expected because the figure set was revised several times. Setting clearer intermediate deadlines for those sections would have helped.

Connection to Education

The thesis draws directly on courses in artificial intelligence, algorithm design, and IoT systems. Concepts from reinforcement learning and Markov Decision Processes that I studied in earlier courses formed the theoretical basis of the algorithm. Work on IoT networks helped me model the battery and sensing environment realistically. The Design Science Research framework was new to me at the start of the project, but it became a useful scaffold for organising the work.

Future Relevance and Use of AI Tools

The experience of building and evaluating a learning-based algorithm under real hardware constraints is directly relevant to my planned career in intelligent systems and embedded software. Understanding how to bridge the gap between simulation and deployment is a practical skill I will carry forward.

Several AI tools were used during the project. Claude (Anthropic) was used to assist with code drafting, $\text{\LaTeX}$ formatting, and editing. GitHub

Copilot helped with Python implementation. All code was reviewed and verified by the authors before use. The research design, analysis, and conclusions were carried out independently. I found these tools most useful for reducing repetitive work, but they required careful checking, especially for citations and technical accuracy.

Overall Satisfaction

I am satisfied with the outcome of the thesis. TQ-IoT achieves what we set out to demonstrate: a configurable energy-freshness trade-off that no fixed rule can replicate. The most practically important finding, that action-cost modelling assumptions strongly affect simulation results, came out clearly and is, I think, a genuinely useful observation for anyone working in this area. With more time, I would have extended the evaluation to multi-node networks and variable channel conditions, which remain the most significant open questions.

Reflection: Peeyush Kelkar

Alignment with the Thesis Course Objectives

The thesis course asks students to demonstrate the ability to carry out qualified academic work, contribute to knowledge, apply scientific methods correctly, and reflect critically on their own work. I believe this thesis addresses these objectives, though with varying degrees of success across different areas.

The main contribution is the TQ-IoT framework, which combines two independent Q-tables with Chebyshev scalarisation to produce a configurable energy-freshness trade-off for IoT sensor scheduling. Implementing and evaluating this framework required sustained independent work over several months, covering algorithm design, simulation engineering, hardware trace processing, and written analysis.

Objectives Met Well

I feel the scientific grounding came together well. The literature review helped me understand how prior work in AoI scheduling and MORL-IoT relate to each other, and where the genuine gap was. Positioning TQ-IoT clearly against prior work, particularly the comparison with Kodu and Zaki in Table 2.1, gave the thesis a sharper contribution statement than it would otherwise have had.

The method application was also handled carefully. The DSR framework gave the project a principled structure, and I made deliberate choices about data collection, data analysis, and the limits of the simulation approach. The decision to evaluate using trace-replay rather than live deployment was a conscious boundary, and I think it was the right one given the available time and infrastructure. Writing up the alternative design choices in Section 3.6 was a useful discipline: it forced me to articulate why we made each decision, not just what we decided.

The sensitivity analysis was one of the more valuable parts of the thesis for me personally. Discovering that action-cost assumptions strongly affect apparent policy performance was not something I anticipated at the start of the project. The measured hardware power values were close enough together that the straightforward approach, normalising directly from OML traces, produced a cost signal too compressed for the agent to learn from. Working through that problem and landing on physically-motivated fallback costs, then validating the sensitivity in the results, turned what could have been a limitation into one of the three main findings. That felt like real research rather than execution.

Objectives Less Fully Met

The objective of contributing to knowledge beyond the existing state of the art was partially met but could have been stronger. TQ-IoT combines existing techniques in a new way and evaluates them against real hardware traces, which is a genuine advance over the closest prior work. But the algorithmic novelty is incremental rather than transformative. A stronger contribution would have involved multi-node coordination or an online preference-adaptation mechanism, both of which remain as future work. I am satisfied that the thesis is honest about this rather than overclaiming.

Time management was also a challenge at certain points. The data processing pipeline for the FIT IoT-Lab traces was more complex than I had anticipated, particularly the alignment of sensor, power, and RSSI logs across nodes and experiments. That work took longer than planned and compressed the time available for the results and discussion chapters, which would have benefited from more revision cycles.

I also found it harder than expected to write the discussion chapter in a way that felt genuinely analytical rather than merely descriptive. The taxonomy of static policy failures in Section 5.2 came together well, but the initial drafts of the limitations section were too brief. Getting the balance right between honest acknowledgement of what the evaluation could not show and a fair account of what it did show required more iteration than I expected.

Connection to Education

Several courses from my programme were directly relevant. Work on reinforcement learning and dynamic programming gave me the theoretical

background for Q-learning and MDPs. Courses on network systems helped me understand the IoT context and the physical constraints of battery-powered nodes. Prior exposure to probability and stochastic processes made the Markov chain model for channel transitions straightforward to reason about.

Design Science Research was a new framework for me at the start of the project. Studying it through the Johannesson and Perjons textbook, and working through how its five phases mapped onto our actual workflow, gave me a clearer sense of how to structure a constructive research project than I had before. I expect to draw on that understanding in future work that involves building and evaluating systems rather than purely analysing existing ones.

Future Relevance and Use of AI Tools

The skills developed during this thesis, building and evaluating a multi-objective learning system, processing real hardware traces, and communicating results clearly in writing, are directly applicable to research and engineering work in the IoT and embedded AI space. I am particularly interested in extending this work toward online adaptation and multi-node settings, and I think the framework we built provides a reasonable foundation for that.

AI tools played a supporting role in the project. Claude was used for code assistance, $\text{\LaTeX}$ editing, and drafting text sections that were then substantially revised. GitHub Copilot helped with repetitive Python implementation tasks. All outputs were reviewed carefully before inclusion. The research design, experimental analysis, and interpretation of results were carried out independently. I found the tools genuinely useful for productivity, but they required consistent vigilance. In particular, I had to correct errors in cited material and in technical claims about the hardware more than once, which reinforced for me that AI-assisted writing requires the same standard of checking as any other source.

Overall Satisfaction

I am broadly satisfied with the thesis. The core result, a configurable trade-off frontier that no static policy can replicate, is demonstrated clearly and consistently across three hardware experiments. The writing improved substantially through revision, and I feel the final version presents the work

honestly, including its limitations. The sensitivity analysis finding is the result I am most pleased with: it was unexpected, it changed how I thought about the problem, and it turned into a practical recommendation that I think is genuinely useful for anyone doing simulation-based IoT policy evaluation.

The main thing I would do differently is plan the results and discussion chapters more carefully from the start. In practice they were treated as a final phase to be written once the experiments were done, but they required as much thinking as the methodology chapters, and they would have been stronger with more time.